

These payoffs can be equivalently summarized as:

	Firm Value $V < \$100$	Firm Value $V > \$100$
Hold a Risk-Free Bond	\$100	\$100
Write a Put with Strike = \$100	$V - \$100$	0
Total	V	\$100

Thus, the payoff to a corporate bondholder is equivalent to the bondholder holding a risk-free bond and selling a put with strike price equal to the face value of the debt. Thus, bondholders have written a put option on the firm’s assets. Corporate bondholders are *short volatility*. Equity holders, conversely, hold a call option on the firm’s assets and are long volatility.

From this decomposition, we can see that

$$\text{Risk-Free Bond} + \text{Put} = \text{Corporate Bond}.$$

Or, rearranging we have

$$\text{Corporate Bond} - \text{Risk-Free Bond} = \text{Short Put Option}. \tag{9.4}$$

The corporate bond spread is hence the value of a short put option on the assets of the borrower with a strike price equal to the face value of the debt. The credit spread increases with volatility, leverage, and maturity.

The intuition behind the fact that the credit spread is a put option is that equity holders have limited liability and are paid second after bond holders. Equity holders have borrowed the par value of debt (\$100 in the above example) from creditors. Equity holders can walk away from their obligations and put the firm to creditors when the firm value is low. It is in this sense that they own a put option. By put–call parity, equity holders have a call option on the firm—equity owners are long volatility just as debt holders are short volatility. Another way to view this is that equity holders get all the upside, which is the call option on the firm’s assets. *Bond holders only have downside risk*, and get none of the upside. Thus, bond holders price in the downside risk, which is the short put position.

Equation (9.4) predicts that credit spreads widen as volatility increases. Figure 9.14 plots VIX volatility, which is a measure of volatility from option prices, with the Baa minus Treasury credit spread (Baa-Tsy) in Panel A and VIX volatility with the Caa minus Treasury credit spread (Caa-Tsy) in Panel B from January 1986 to December 2011. In each panel, VIX is plotted on the left-hand axis and the credit spreads are plotted on the right-hand axis. The correlations between volatility and credit spreads are high: 64% for Baa-Tsy and 56% for Caa-Tsy. Thus, volatility is a crucial factor for corporate bonds, and high volatility occurs contemporaneously with high credit spreads and lower realized returns. Note that in both panels in Figure 9.14, the temporary jump in volatility due to the 1987 crash was not